

人人媒好 - 推荐码管理系统流程图

一、完整业务流程图

```
flowchart TD
    Start([Start]) --> InputCode{InputCode{}}
    InputCode --> VerifyCode{VerifyCode{}}
    InputCode --> Register[Register]
    VerifyCode --> BindUser[BindUser]
    VerifyCode --> ShowError[ShowError]
    BindUser --> UpdateCode[UpdateCode]
    UpdateCode --> UpdateUser[UpdateUser]
    UpdateUser --> Register
    Register --> UserRole{UserRole{}}
    UserRole --> NormalUser[NormalUser]
    UserRole --> ApplyPage[ApplyPage]
    ApplyPage --> ChooseType[ChooseType]
    ChooseType --> SubmitApply[SubmitApply]
    SubmitApply --> AdminReview[AdminReview]
    AdminReview --> AssignCode[AssignCode]
    AdminReview --> NotifyReject[NotifyReject]
    AssignCode --> BindCodeToUser[BindCodeToUser]
    BindCodeToUser --> ActivateReferrer[ActivateReferrer]
    ActivateReferrer --> ReferrerDashboard[ReferrerDashboard]
    ReferrerDashboard --> DevelopUsers[DevelopUsers]
    DevelopUsers --> NewUserReg[NewUserReg]
    NewUserReg --> BindUser
    ReferrerDashboard --> BindSubReferrer[BindSubReferrer]
    BindSubReferrer --> CheckSubCode{CheckSubCode{}}
    CheckSubCode --> CheckNotBound{CheckNotBound{}}
    CheckSubCode --> Error1[Error1]
    CheckNotBound --> BindRelationship[BindRelationship]
    CheckNotBound --> Error2[Error2]
    BindRelationship --> UpdateReferredBy[UpdateReferredBy]
    UpdateReferredBy --> LogRelationship[LogRelationship]
    LogRelationship --> CalcCommission[CalcCommission]
    CalcCommission --> CommissionRecord[CommissionRecord]
    CommissionRecord --> CommissionStatus{CommissionStatus{}}
    CommissionStatus --> pending[pending]
    CommissionStatus --> confirmed[confirmed]
    CommissionStatus --> paid[paid]
    pending --> WaitConfirm[WaitConfirm]
    confirmed --> ReadyPay[ReadyPay]
    paid --> AlreadyPaid[AlreadyPaid]
    WaitConfirm --> AdminConfirm[AdminConfirm]
    AdminConfirm --> ReadyPay
    ReadyPay --> WechatPay[WechatPay]
    WechatPay --> NotifyPaid[NotifyPaid]
    ReferrerDashboard --> ViewTeam[ViewTeam]
    ViewTeam --> ShowTree[ShowTree]
    classDef success fill:#d4edda,stroke:#28a745
    classDef error fill:#f8d7da,stroke:#dc3545
    classDef process fill:#d1ecf1,stroke:#17a2b8
    class BindUser,UpdateCode,UpdateUser,ActivateReferrer,BindRelationship success
    class ShowError,Error1,Error2 error
    class AdminReview,AdminConfirm,WechatPay process
```

二、推荐码生命周期图

stateDiagram-v2

```

[*] --> ■■■: Admin ■■■■
■■■ --> ■■■: ■■■■
■■■ --> ■■■: ■■■■■■
■■■ --> ■■■: ■■■■■■
■■■ --> ■■■: use_count >= max_uses
■■■ --> ■■■: ■■■ expires_at
■■■ --> ■■■: Admin ■■■■
■■■ --> [*]
■■■ --> [*]
■■■ --> [*]

note right of ■■■
    ■■■■■■
    ■■■■■■■■
end note

note right of ■■■
    referrer_id ■■■
    ■■■■■■■■
end note

note right of ■■■
    ■■■■■■■■■■
    use_count ■■■■
end note

```

三、数据库表关系图

```

erDiagram
    users ||--o{ referral_codes : "■■■(referrer_id)"
    users ||--o{ referral_usage_logs : "■■■■■"
    users ||--o{ referral_relationships : "■■■■■(referrer_id)"
    users ||--o{ referral_relationships : "■■■■■(referee_id)"
    users ||--o{ commission_records : "■■■■■(user_id)"

    referral_codes ||--o{ referral_usage_logs : "■■■■(code)"
    referral_codes {
        int id PK
        string code UK "■■■■■■■"
        string code_type "■■:creator/public_welfare/..."
        int referrer_id FK "■■■■■ID"
        string referred_by_code "■■■■■"
        string status "active/inactive/depleted"
        int use_count "■■■■■"
        int max_uses "■■■■■■■"
        string expires_at "■■■■■"
        string batch_id "■■■ID"
        text remark "■■■"
    }

    referral_usage_logs {
        int id PK
        string code FK "■■■■"
        int user_id FK "■■■■■ID"
        string scene "■■:register/manual_bind"
        string created_at
    }

    referral_relationships {
        int id PK
        int referrer_id FK "■■■ID"
        string referrer_code "■■■■■■■"
        int referee_id FK "■■■■■ID"
        string referee_code "■■■■■■■"
        string status "active/inactive"
        int level "■■:1=■■,2=■■"
        decimal total_commission "■■■"
        text remark
    }

    commission_records {
        int id PK
        int user_id FK "■■■ID"
        int referral_relationship_id FK "■■■■■■■"
        string type "■■:register/upgrade/purchase"
        decimal amount "■■■■■"
        string status "pending/confirmed/paid"
        string paid_at "■■■■■"
    }

```

```

user_referrals {
    int id PK
    int referrer_id FK
    int referee_id FK
    string referral_code
    string status
}

```

四、推荐关系绑定时序图

```

sequenceDiagram
    participant Admin as 用户
    participant Frontend as 前端
    participant Backend as 后端API
    participant DB as 数据库

    Admin->>Frontend: 1. 选择推荐码
    Frontend->>Backend: GET /api/admin/referral-codes/list
    Backend->>DB: SELECT * FROM referral_codes
    DB-->>Backend: 返回数据
    Backend-->>Frontend: {code:0, data: [...]}
    Frontend->>Frontend: 显示Picker

    Admin->>Frontend: 2. 绑定推荐关系
    Admin->>Frontend: 3. 提交

    Frontend->>Frontend: 4. 提交
    Frontend->>Backend: POST /api/admin/referral-codes/bind
    Note over Frontend,Backend: {referrer_code: "LCRG48KL", referred_code: "GYRG..."}

    Backend->>Backend: 5. 校验
    Backend->>DB: SELECT * FROM referral_codes WHERE code = ?
    DB-->>Backend: 返回数据

    alt 校验失败
        Backend-->>Frontend: {code:-1, message:"校验失败"}
        Frontend-->>Admin: 提示
    end

    Backend->>DB: SELECT * FROM referral_codes WHERE code = ?
    DB-->>Backend: 返回数据

    alt 校验成功
        Backend-->>Frontend: {code:-1, message:"校验成功"}
        Frontend-->>Admin: 提示
    end

    Backend->>DB: CHECK referred_by_code IS NOT NULL

    alt 已存在
        Backend-->>Frontend: {code:-1, message:"已存在"}
        Frontend-->>Admin: 提示
    end

    Backend->>DB: BEGIN TRANSACTION
    DB-->>Backend: 开始事务

    Backend->>DB: UPDATE referral_codes SET referred_by_code = ? WHERE code = ?
    DB-->>Backend: 更新成功

    Backend->>DB: INSERT INTO referral_relationships (...)
    DB-->>Backend: 插入成功

    Backend->>DB: COMMIT
    DB-->>Backend: 提交成功

    Backend-->>Frontend: {code:0, message:"绑定成功"}
    Frontend-->>Admin: 提示
    Frontend->>Frontend: 结束

```

五、佣金计算流程图

```

flowchart TD
    Start([开始]) --> TriggerType{触发类型}
    TriggerType --> Calc1[计算佣金1]
    TriggerType --> Calc2[计算佣金2]

```

```

TriggerType -->|■■■■■■| Calc3[■■■■■■]
TriggerType -->|■■| Calc4[■■■■■■]

Calc1 --> GetReferral[■■■■■■]
Calc2 --> GetReferral
Calc3 --> GetReferral
Calc4 --> GetReferral

GetReferral --> CheckLevel{■■■■■?}

CheckLevel -->|■■■■■| Rate1[■■20%]
CheckLevel -->|■■■■■| Rate2[■■15%]
CheckLevel -->|■■■■■| Rate3[■■10%]
CheckLevel -->|■■■■■| Rate4[■■12%]
CheckLevel -->|■■■■■| Rate5[■■25%]

Rate1 --> CheckSub[■■■■■■]
Rate2 --> CheckSub
Rate3 --> CheckSub
Rate4 --> CheckSub
Rate5 --> CheckSub

CheckSub -->|■■■| CalcSubCommission[■■■■■■]
CheckSub -->|■■■| GenerateRecord[■■■■■■]

CalcSubCommission --> SubRate{■■■■}
SubRate -->|■■| SubRate1[■■5%]
SubRate -->|■■| SubRate2[■■3%]
SubRate -->|■■| SubRate3[■■2%]

SubRate1 --> GenerateRecord
SubRate2 --> GenerateRecord
SubRate3 --> GenerateRecord

GenerateRecord --> InsertCommission[INSERT INTO commission_records]
InsertCommission --> NotifyReferrer[■■■■■■■■]
NotifyReferrer --> End([■■])

classDef trigger fill:#fff3cd,stroke:#ffc107
classDef calculate fill:#d1ecf1,stroke:#17a2b8
classDef notify fill:#d4edda,stroke:#28a745

class Start,TriggerType trigger
class Calc1,Calc2,Calc3,Calc4,CheckLevel,Rate1,Rate2,Rate3,Rate4,Rate5,Check...
class NotifyReferrer,End notify

```

六、关键业务规则

1. 推荐码生成规则

- **公益推荐官**：前缀 `GYRG` + 4位随机字符
- **联创推荐官**：前缀 `LCRG` + 4位随机字符
- **专业推荐官**：前缀 `ZYRG` + 4位随机字符
- **社区服务站**：前缀 `SQZD` + 4位随机字符
- **城市合伙人**：前缀 `CSHH` + 4位随机字符

2. 推荐关系层级

- **Level 1**：直接推荐（A 推荐 B）
- **Level 2**：间接推荐（A 推荐 B，B 推荐 C，A 获得 C 的间接奖励）

3. 佣金结算周期

- **T+1**：用户注册/升级触发佣金（pending）
- **T+7**：管理员确认无异常后（confirmed）

- ****每月15日****：批量发放上月 confirmed 佣金 (paid)

4. 防作弊机制

- 同一 IP 24小时内只能使用 3 次推荐码
- 推荐关系建立后 7 天内不能解除
- 退款订单对应的佣金自动扣除

— — —

附录：快速 SQL 查询

查询推荐官的完整团队

```

WITH RECURSIVE team AS (
  -- ██████████
  SELECT
    rr.referee_id,
    rr.referrer_code,
    rr.referee_code,
    rr.level,
    1 as depth
  FROM referral_relationships rr
  WHERE rr.referrer_id = ? -- █████ID

  UNION ALL

  -- ████████
  SELECT
    rr.referee_id,
    rr.referrer_code,
    rr.referee_code,
    rr.level,
    t.depth + 1
  FROM referral_relationships rr
  INNER JOIN team t ON rr.referrer_id = t.referee_id
  WHERE t.depth < 5 -- ███5█
)
SELECT * FROM team;

```

统计推荐官业绩

```
SELECT
    u.id,
    u.nickname,
    rc.code as referral_code,
    COUNT(DISTINCT rr.referee_id) as total_referees,
    COUNT(DISTINCT CASE WHEN u_reg.created_at >= date('now', '-30 days') THEN rr...
    COALESCE(SUM(cr.amount), 0) as total_commission
FROM users u
LEFT JOIN referral_codes rc ON rc.referrer_id = u.id
LEFT JOIN referral_relationships rr ON rr.referrer_id = u.id
LEFT JOIN commission_records cr ON cr.user_id = u.id AND cr.status = 'paid'
WHERE u.role IN ('creator', 'professional', 'public_welfare')
GROUP BY u.id;
```